CV ASSIGNMENT-1

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1. Objective

The objective of this assignment is to construct a unified panoramic image by stitching multiple overlapping photographs. The task involves determining geometric relationships between images, aligning them into a common coordinate system, and ensuring visual consistency. A comparison is performed between a naïve stitching approach and a refined stitching pipeline.

2. Input Data

Three overlapping images were captured using a mobile phone camera. The images were taken by rotating the camera around a fixed point to approximately maintain the optical center. Care was taken to ensure sufficient overlap (about 30–40%) between adjacent images, which is necessary for reliable feature matching.

3. Methodology

The panorama was generated using a classical feature-based image stitching pipeline. Each stage of the pipeline is described below.

Step 1: Image Loading

The input images were loaded using OpenCV. Assertions were used to verify successful image loading, preventing downstream errors caused by incorrect file paths or missing images.

Purpose:
To import image data into the processing pipeline.

Step 2: Grayscale Conversion

All images were converted to grayscale before feature detection. Feature extraction algorithms operate on intensity patterns rather than color, making grayscale images sufficient and computationally efficient.

Purpose:
To simplify feature detection by removing color dependencies.

Step 3: Feature Detection using SIFT

Scale-Invariant Feature Transform (SIFT) was used to detect distinctive keypoints and compute descriptors. SIFT provides invariance to scale and rotation, making it highly suitable for panorama stitching.

Purpose:
To identify repeatable and distinctive visual structures across images.

Outcome:
A set of keypoints and descriptor vectors for each image.

Step 4: Feature Matching

Feature correspondence between overlapping images was established using a Brute Force Matcher with K-Nearest Neighbors (KNN). Lowe's Ratio Test was applied to eliminate ambiguous matches.

Purpose:
To determine reliable point correspondences between images.

Why Ratio Test:
Raw descriptor matching produces incorrect matches. The ratio test ensures only confident matches are retained.

Step 5: Homography Estimation with RANSAC

Homography matrices were computed from matched keypoints using RANSAC. RANSAC rejects outliers and estimates a robust geometric transformation.

Purpose:
To model the projective transformation required for image alignment.

Why Homography:
Panoramic images involve perspective distortions that cannot be handled by simple translations.

Step 6: Panorama Geometry Computation

The valid spatial extent of the panorama was determined by projecting image corners through homography matrices. A translation matrix was computed to avoid negative coordinates.

Purpose:
To define a common coordinate frame for all warped images.

Step 7: Image Warping

Images were projected into the panorama coordinate frame using perspective warping. The center image served as the reference frame.

Purpose:
To geometrically align images under perspective transformations.

Outcome:
Spatially aligned image layers.

Naive Stitch (No Blending)

Step 8: Blending

Simple averaging blending was applied in overlapping regions. Overlapping pixel values were averaged to reduce visible seams and intensity discontinuities.

Purpose:
To improve visual smoothness and reduce stitching artifacts.

Output:

Step 9: Final Cleanup and Cropping

Black canvas regions introduced during warping were removed through automatic cropping based on valid pixel regions.

Purpose:
To produce a clean rectangular panorama suitable for presentation.

OUTPUT:

Final Panorama

4. Naive Stitching Approach

For comparison, a naïve stitching method was implemented. This method directly overlays images without estimating geometric transformations or blending.

Naive Stitch Characteristics

- No geometric alignment
- Hard seams between images

- Visible misalignment
- Distorted structures
- Poor visual quality

Reason for Failure

Images captured from different viewpoints undergo perspective distortions. Direct overlay ignores these distortions, leading to incorrect alignment.

5. Refined Stitching Pipeline

The refined pipeline integrates feature detection, robust matching, homography estimation, perspective warping, blending, and cropping.

Advantages

- Accurate geometric alignment
- Outlier rejection via RANSAC
- Reduced seam artifacts
- Improved visual consistency
- Structurally correct panorama

6. Comparison of Results

Criterion	Naive Stitch	Final Panorama
Alignment	Incorrect	Geometrically correct
Seams	Clearly visible	Smooth transitions
Artifacts	Severe distortions	Minimal artifacts
Visual Quality	Poor	Significantly improved

Observations

The naive method results in visible discontinuities and structural misalignment. The refined pipeline successfully aligns images and produces a coherent panoramic image.

7. Conclusion

This assignment demonstrates that panorama stitching requires both geometric and photometric processing. Feature-based matching combined with homography estimation enables accurate alignment, while blending and cropping improve visual quality. The comparison clearly illustrates the limitations of naïve overlay and the effectiveness of the proposed stitching pipeline.